INDIAN DATA CLUB

Day 21: Query Organisation (CTEs)

🎯 ***Objective***

To simplify complex SQL queries using the WITH clause and CTEs for better readability, organization, and structured analysis.

🔍 ***Topics Covered***

- **WITH clause**
- **CTEs (Common Table Expressions)**

WITH Clause / CTEs (Common Table Expressions)

The **WITH clause** is used to create temporary result sets that exist only for that query.
☞ Think of it as creating a **temporary named table** for better readability

Syntax:

```
WITH temp_table AS (
    SELECT ...
)
SELECT * FROM temp_table;
```

```
WITH cte1 AS (
    SELECT ...
),
cte2 AS (
    SELECT ...
)
SELECT *
FROM cte1
JOIN cte2 ON ...
```

Example

Find the average satisfaction of each service and then return only those above 80.

```
WITH service_stats AS (
    SELECT
        service_id,
        AVG(satisfaction) AS avg_satisfaction
    FROM patient_feedback
    GROUP BY service_id
)
SELECT *
FROM service_stats
WHERE avg_satisfaction > 80;
```

Practice Questions:

```sql
-- Create a CTE to calculate service statistics, then query from it.
WITH service_stats AS (
    SELECT
        service,
        ROUND(AVG(patient_satisfaction),1) AS avg_satisfaction,
        SUM(patients_admitted) AS total_admissions,
        COUNT(*) AS weeks_count
    FROM services_weekly
    GROUP BY service
)

SELECT
    service,
    avg_satisfaction,
    total_admissions,
    weeks_count
FROM service_stats;
```

service	avg_satisfaction	total_admissions	weeks_count
emergency	77.9	1185	52
surgery	79.3	1686	52
general_medicine	81.2	2332	52
ICU	81.6	648	52

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Practice Questions:

```sql
-- Use multiple CTEs to break down a complex query into logical steps.
WITH
-- Step 1: Weekly admissions summary
weekly_admissions AS (
    SELECT
        week,
        service,
        patients_admitted,
        patients_refused
    FROM services_weekly
),
-- Step 2: Weekly staff presence for each service
weekly_staff AS (
    SELECT
        week,
        service,
        COUNT(*) AS staff_present
    FROM staff_schedule
    WHERE present = 1
    GROUP BY week, service
),
-- Step 3: Combine weekly admissions + staff info
combined AS (
    SELECT
        a.week,
        a.service,
        a.patients_admitted,
        a.patients_refused,
        s.staff_present
    FROM weekly_admissions a
    LEFT JOIN weekly_staff s
        ON a.week = s.week
        AND a.service = s.service
```

```sql
-- Final Output
SELECT *
FROM combined
ORDER BY service, week;
```

week	service	patients_admitted	patients_refused	staff_present
1	emergency	32	44	35
2	emergency	28	141	37
3	emergency	32	145	NULL
4	emergency	32	125	36
5	emergency	25	363	35
6	emergency	34	164	NULL
7	emergency	33	94	35
8	emergency	26	214	36
9	emergency	28	85	NULL
10	emergency	17	113	35
11	emergency	16	81	34
12	emergency	28	319	NULL
13	emergency	23	119	34
14	emergency	15	82	29
15	emergency	17	36	NULL
16	emergency	23	121	36
17	emergency	21	112	37
18	emergency	18	82	NULL
19	emergency	16	54	35
20	emergency	21	24	37

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Practice Questions:

```sql
-- Build a CTE for staff utilization and join it with patient data.
WITH staff_utilization AS (
    SELECT
        s.service,
        s.staff_id,
        s.staff_name,
        COUNT(p.patient_id) AS total_patients_assigned
    FROM staff s
    LEFT JOIN patients p
        ON s.service = p.service
    GROUP BY s.service, s.staff_id, s.staff_name
)
SELECT
    p.patient_id,
    p.name AS patient_name,
    p.service,
    su.staff_id,
    su.staff_name,
    su.total_patients_assigned
FROM patients p
LEFT JOIN staff_utilization su
    ON p.service = su.service
ORDER BY su.service DESC;
```

patient_id	patient_name	service	staff_id	staff_name	total_patients_assigned
PAT-f83eb12d	Patricia Johnson	surgery	STF-3d6f6916	Philip Cannon	254
PAT-f83eb12d	Patricia Johnson	surgery	STF-42844552	Shane Henderson	254
PAT-f83eb12d	Patricia Johnson	surgery	STF-46e008d1	Tricia Valencia	254
PAT-f83eb12d	Patricia Johnson	surgery	STF-5d51a8ea	Eric Carney	254
PAT-f83eb12d	Patricia Johnson	surgery	STF-7a5180f0	Fred Smith	254
PAT-f83eb12d	Patricia Johnson	surgery	STF-b19d1900	Jessica Holmes	254
PAT-f83eb12d	Patricia Johnson	surgery	STF-bc33b33b	Cassandra Gaines	254
PAT-f83eb12d	Patricia Johnson	surgery	STF-d7c28217	Debra Davidson	254
PAT-f83eb12d	Patricia Johnson	surgery	STF-f2a7bd6f	Nathan Maldonado	254
PAT-f93d4260	Barbara Hester	surgery	STF-f2a7bd6f	Nathan Maldonado	254
PAT-feaf8fa9	Joshua Reed	surgery	STF-0aaed714	Rebecca Henderson	254
PAT-fab8ee29	Dustin Gallegos	surgery	STF-00fbd582	John Pierce	254
PAT-fab8ee29	Dustin Gallegos	surgery	STF-0aaed714	Rebecca Henderson	254
PAT-fab8ee29	Dustin Gallegos	surgery	STF-13f243f8	Paula Moreno	254
PAT-fab8ee29	Dustin Gallegos	surgery	STF-15269c02	Sherry Decker	254
PAT-fab8ee29	Dustin Gallegos	surgery	STF-15c07995	Anthony Rodriguez	254
PAT-fab8ee29	Dustin Gallegos	surgery	STF-177e5df8	Angelica Tucker	254
PAT-fab8ee29	Dustin Gallegos	surgery	STF-1ded4330	Joshua Blair	254
PAT-fab8ee29	Dustin Gallegos	surgery	STF-2bfe24c1	Jeffrey Chavez	254
PAT-fab8ee29	Dustin Gallegos	surgery	STF-2c155ebf	Elizabeth Fowler	254

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Daily Challenge:

```sql
-- Create a comprehensive hospital performance dashboard using CTEs.
-- Calculate: 1) Service-level metrics (total admissions, refusals, avg satisfaction),
-- 2) Staff metrics per service (total staff, avg weeks present),
-- 3) Patient demographics per service (avg age, count).
-- Then combine all three CTEs to create a final report showing service name, all calculated metrics,
-- and an overall performance score (weighted average of admission rate and satisfaction).
-- Order by performance score descending
```

```sql
WITH service_metrics AS (
    SELECT
        service,
        SUM(patients_admitted) AS total_admissions,
        SUM(patients_refused) AS total_refusals,
        ROUND(AVG(patient_satisfaction), 1) AS avg_satisfaction,
        CASE
            WHEN SUM(patients_admitted + patients_refused) = 0 THEN 0
            ELSE ROUND(SUM(patients_admitted) * 1.0 / (SUM(patients_admitted + patients_refused)) * 100,
        END AS admission_rate
    FROM services_weekly
    GROUP BY service
),
```

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Daily Challenge:

```sql
staff_weeks AS (
    SELECT
        s.service,
        s.staff_id,
        COUNT(ss.week) AS weeks_present
    FROM staff s
    LEFT JOIN staff_schedule ss
        ON s.staff_id = ss.staff_id AND ss.present = 1
    GROUP BY s.service, s.staff_id
),
staff_metrics AS (
    SELECT
        service,
        COUNT(staff_id) AS total_staff,
        ROUND(AVG(weeks_present), 1) AS avg_weeks_present
    FROM staff_weeks
    GROUP BY service
),
patient_demo AS (
    SELECT
        service,
        ROUND(AVG(age), 1) AS avg_patient_age,
        COUNT(patient_id) AS patient_count
    FROM patients
    GROUP BY service
)
```

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Daily Challenge:

```sql
SELECT
    sm.service,
    sm.total_admissions,
    sm.total_refusals,
    sm.avg_satisfaction,
    sm.admission_rate,
    st.total_staff,
    st.avg_weeks_present,
    pd.avg_patient_age,
    pd.patient_count,
    ROUND((sm.admission_rate + sm.avg_satisfaction) / 2, 1) AS performance_score
FROM service_metrics sm
LEFT JOIN staff_metrics st ON sm.service = st.service
LEFT JOIN patient_demo pd ON sm.service = pd.service
ORDER BY performance_score DESC;
```

service	total_admissions	total_refusals	avg_satisfaction	admission_rate	total_staff	avg_weeks_present	avg_patient_age	patient_count	performance_score
ICU	648	141	81.6	82.1	48	31.2	44.2	241	81.9
surgery	1686	555	79.3	75.2	22	31.7	45.6	254	77.3
general_medicine	2332	1938	81.2	54.6	27	30.7	46.2	242	67.9
emergency	1185	5008	77.9	19.1	29	31.3	45.3	263	48.5